

Self-Supervised Image Denoising: A Comparative Study of Single-Image Methods

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Abstract

Image denoising is a fundamental problem in low-level computer vision, requiring the recovery of a clean image from its noisy observation. Supervised deep learning methods achieve state-of-the-art performance but require large paired or unpaired datasets. In this report, we evaluate three self-supervised and unsupervised denoising methods — Deep Image Prior (DIP), Self2Self, and Neighbor2Neighbor — on 16 images from the CBSD68 benchmark under additive white Gaussian noise with $\sigma = 25$. Each method is evaluated in a single-image setting, and Neighbor2Neighbor is additionally evaluated in a pretrained setting to establish an upper-bound reference. Results are reported in terms of Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), edge preservation, and inference time. The pretrained Neighbor2Neighbor achieves the best overall performance at 30.56 dB PSNR and 0.883 SSIM, while among single-image methods, Deep Image Prior achieves the highest PSNR (28.44 dB) and Self2Self achieves the highest SSIM (0.821) at significantly lower inference time. These findings highlight the trade-off between denoising quality, computational cost, and reliance on external training data.

1 Introduction

Image denoising is the recovery of a clean image $\mathbf{x}$ from a noisy observation $\mathbf{y} = \mathbf{x} + \boldsymbol{\epsilon}$. It is one of the most fundamental and well-studied problems in image processing. Noise corrupts images during acquisition across a wide range of domains, from digital photography and medical imaging to remote sensing and scientific microscopy. Producing an accurate and visually faithful denoised image remains a challenging ill-posed inverse problem.

Traditional methods such as Non-Local Means [Buades et al., 2005] and BM3D [Dabov et al., 2007] exploit handcrafted image priors — notably, the statistical self-similarity of natural image patches — to achieve strong denoising performance without any learned parameters. The advent of deep learning brought supervised methods such as DnCNN [Zhang et al., 2017] and FFDNet [Zhang et al., 2018], which dramatically improved performance by learning rich image priors from large datasets of clean/noisy image pairs.

However, supervised methods depend critically on the availability of clean training data, which is difficult or impossible to obtain in many practical settings. This has motivated a growing body of research into *self-supervised* and *unsupervised* denoising, where the

model learns from noisy observations alone. The Noise2Noise framework [Lehtinen et al., 2018] showed that a network can be trained using pairs of noisy images without clean targets. Subsequent work pushed further toward single-image denoising: Noise2Void [Krull et al., 2019] and Noise2Self [Batson and Royer, 2019] introduced blind-spot strategies, Self2Self [Quan et al., 2020] leveraged Bernoulli dropout masking, Deep Image Prior (DIP) [Ulyanov et al., 2018] exploited the implicit regularisation of a randomly initialised network architecture, and Neighbor2Neighbor [Huang et al., 2021] generated sub-image training pairs from spatially adjacent pixels.

This report presents a systematic empirical comparison of three representative methods — DIP, Self2Self, and Neighbor2Neighbor — evaluated on a subset of the CBSD68 benchmark. We examine both a purely single-image setting (where no external data is used at any stage) and a pretrained setting for Neighbor2Neighbor to quantify the gap between data-free and data-driven approaches. Beyond standard PSNR and SSIM metrics, we also report edge preservation and per-image inference time to characterise the practical trade-offs between these methods.

2 Background

2.1 Problem Definition

Let $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ denote a clean image corrupted by additive white Gaussian noise (AWGN):

$$\mathbf{y} = \mathbf{x} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \quad (1)$$

where σ is the noise standard deviation. The goal of denoising is to recover an estimate $\hat{\mathbf{x}}$ of the clean image from the noisy observation $\mathbf{y}$ alone.

From a Bayesian perspective, the optimal estimate under mean squared error loss is the posterior mean:

$$\hat{\mathbf{x}} = \arg \max_{\mathbf{x}} p(\mathbf{x} \mid \mathbf{y}) = \arg \max_{\mathbf{x}} p(\mathbf{y} \mid \mathbf{x}) p(\mathbf{x}), \quad (2)$$

where $p(\mathbf{y} \mid \mathbf{x})$ is the Gaussian likelihood from Equation (1) and $p(\mathbf{x})$ is the image prior encoding statistical regularities of natural images. The central challenge of denoising is the design or learning of an effective image prior.

In the self-supervised and single-image settings considered in this report, no clean images are available at any stage of training or inference. The methods must therefore rely on structural priors implicit in the network architecture, statistical properties of natural images, or consistency constraints derivable from the noisy input alone.

2.2 Classical Methods

Non-Local Means (NLM). Buades et al. [2005] proposed one of the earliest principled denoising methods based on non-local self-similarity. Each pixel is restored as a weighted average of all pixels in the image whose local neighbourhoods are similar, exploiting the recurring patch structure of natural images. While effective on textures, NLM is computationally expensive and tends to over-smooth fine structures.

BM3D. Block-Matching 3D [Dabov et al., 2007] remains among the most competitive classical denoisers. It groups similar 2D image patches into 3D arrays, applies a sparsifying

transform in both the spatial and grouping dimensions, performs Wiener filtering, and aggregates the filtered patches back. BM3D exploits both the transform-domain sparsity of natural images and their non-local self-similarity, achieving a strong balance between noise suppression and detail preservation.

Supervised Deep Denoisers. DnCNN [Zhang et al., 2017] introduced residual learning for image denoising, training a deep CNN to predict the noise residual rather than the clean image directly. FFDNet [Zhang et al., 2018] extended this to a flexible framework accepting the noise level as an input, enabling a single model to handle a range of noise intensities. These methods achieve significant improvements over classical approaches but require large clean training datasets.

2.3 Datasets

CBSD68. The Color Berkeley Segmentation Dataset 68 [Martin et al., 2001] is a standard benchmark for colour image denoising comprising 68 natural images of size 481×321 pixels (or 321×481). Images span diverse content including textures, objects, animals, and scenes. It is the colour counterpart of the widely used BSD68 grayscale benchmark and is the standard evaluation set for comparing denoising methods at noise levels $\sigma \in \{15, 25, 50\}$. In this study, all images are corrupted with AWGN at $\sigma = 25$, yielding an average noisy PSNR of approximately 20.59 dB.

For this evaluation, a representative subset of 16 images is used to allow the computationally expensive single-image optimisation methods to be evaluated within feasible time constraints.

3 Method

We evaluate three methods spanning different paradigms of self-supervised and unsupervised denoising. All single-image methods are run independently on each test image with no information shared across images. The pretrained Neighbor2Neighbor variant is included as a reference to quantify the gap between single-image and dataset-trained approaches.

3.1 Deep Image Prior (DIP)

Deep Image Prior [Ulyanov et al., 2018] exploits the observation that an untrained convolutional network acts as an implicit regulariser for natural images. Rather than learning from data, DIP fits a randomly initialised U-Net directly to a single noisy image:

$$\hat{\theta} = \arg \min_{\theta} \|f_{\theta}(\mathbf{z}) - \mathbf{y}\|^2, \quad (3)$$

where $\mathbf{z}$ is a fixed random noise input and f_{θ} is the network. Because the network learns low-frequency image content before fitting high-frequency noise, early stopping at the right iteration yields a denoised output. DIP requires no training data but is sensitive to the choice of stopping criterion and incurs high per-image optimisation cost.

3.2 Self2Self

Self2Self [Quan et al., 2020] generates self-supervised training pairs from a single noisy image using Bernoulli dropout masks. A random binary mask $\mathbf{m} \sim \text{Bernoulli}(p)$ is applied to the input, and the network is trained to predict the masked pixels from the unmasked ones:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{m}} [\| (f_{\theta}(\mathbf{m} \odot \mathbf{y}) - \mathbf{y}) \odot (1 - \mathbf{m}) \|^2] . \quad (4)$$

At inference, predictions are averaged over a large number of independently sampled masks to reduce variance. Self2Self provably converges to the minimum mean squared error estimate under mild conditions and requires no external data.

3.3 Neighbor2Neighbor

Neighbor2Neighbor [Huang et al., 2021] constructs training pairs from a single noisy image by sampling spatially disjoint sub-images from 2×2 pixel neighbourhoods. For each 2×2 block, two diagonally opposite pixels form one training pair, while the other two form a regularisation pair. This produces an implicit Noise2Noise training objective [Lehtinen et al., 2018] without requiring any additional noisy observations.

In the *single* setting, the model is trained from scratch on each test image. In the *pretrained* setting, a single model is trained on a large dataset (ImageNet) and applied directly at inference time, taking approximately 1.5 seconds per image on average.

3.4 Experimental Setup

All experiments were conducted on a machine equipped with an Intel Core i5-9300H CPU (4 cores, 8 threads, 2.40 GHz), 8 GB of system RAM, and an NVIDIA GeForce GTX 1650 GPU with 4 GB of VRAM.

To ensure a fair comparison across methods, all single-image models are evaluated under a protocol which caps the number of gradient updates at **6,000** per image regardless of the method. This constraint prevents any single method from benefiting disproportionately from extended optimisation and ensures that observed performance differences reflect the quality of each method’s training objective and inductive biases rather than differences in compute budget. The pretrained Neighbor2Neighbor variant is exempt from this constraint, as it performs no per-image optimisation.

4 Results

4.1 Evaluation Metrics

We report the following metrics:

- **PSNR** (dB): Peak Signal-to-Noise Ratio between the denoised and clean images. Higher is better.
- **SSIM**: Structural Similarity Index [Zhang et al., 2017], measuring perceptual similarity in terms of luminance, contrast, and structure. Range $[0, 1]$; higher is better.
- **Edge Preservation**: Correlation between edge maps of the denoised and clean images, measuring fidelity to fine structural detail. Range $[0, 1]$; higher is better.

- **Inference Time** (s): Wall-clock time per image on a single GPU.

4.2 Quantitative Results

Table 1 reports average metrics across all 16 test images for each method and mode. Table 2 presents the per-image PSNR breakdown.

Table 1: Average results across 16 CBSD68 images ($\sigma = 25$). **Bold** indicates best among single-image methods; underline indicates overall best.

Method	Mode	PSNR (dB) $\uparrow$	SSIM $\uparrow$	Edge Pres. $\uparrow$	Time (s) $\downarrow$
Noisy input	—	20.59	—	—	—
Deep Image Prior	single	28.44	0.803	0.910	3305.8
Self2Self	single	28.14	0.821	0.893	1751.5
Neighbor2Neighbor	single	27.17	0.782	0.930	3204.4
Neighbor2Neighbor	pretrained	<u>30.56</u>	<u>0.883</u>	<u>0.943</u>	<u>1.51</u>

Table 2: Per-image PSNR (dB) results on 16 CBSD68 images ($\sigma = 25$).

Image	DIP	Self2Self	N2N (single)	N2N (pretrained)
0000	35.67	36.48	33.46	<u>38.41</u>
0001	28.29	28.05	27.76	<u>29.76</u>
0002	31.29	31.44	31.33	<u>32.98</u>
0003	28.04	27.57	24.76	<u>30.11</u>
0004	27.82	27.56	27.92	<u>29.97</u>
0005	28.07	27.94	26.16	<u>30.43</u>
0006	28.08	26.85	24.50	<u>30.10</u>
0007	25.62	25.02	26.31	<u>27.17</u>
0008	25.85	25.35	24.76	<u>27.76</u>
0009	27.38	27.01	26.77	<u>29.28</u>
0010	27.93	27.51	25.95	<u>31.00</u>
0011	29.60	29.61	27.78	<u>32.58</u>
0012	26.72	25.56	24.23	<u>28.31</u>
0013	30.46	30.35	28.72	<u>32.28</u>
0014	28.03	28.00	27.60	<u>30.49</u>
0015	26.24	25.96	26.66	<u>28.40</u>
Average	28.44	28.14	27.17	<u>30.56</u>

4.3 Analysis

Single-image methods. Among the three single-image methods, DIP achieves the highest average PSNR at 28.44 dB, outperforming Self2Self by 0.30 dB and Neighbor2Neighbor (single) by 1.27 dB. However, Self2Self achieves the highest average SSIM (0.821), suggesting it better preserves perceptual structure even when its pixel-level fidelity is marginally lower. Neighbor2Neighbor in the single-image setting achieves the best edge preservation

(0.930) despite its lower PSNR and SSIM, indicating that its sub-image pair strategy is particularly effective at retaining structural boundaries.

Inference time. Self2Self is the most efficient single-image method at an average of 1751.5 seconds per image, approximately half the time of DIP (3305.8 s) and Neighbor2Neighbor single (3204.4 s). All single-image methods are, however, several orders of magnitude slower than the pretrained Neighbor2Neighbor, which runs in an average of 1.51 seconds — approximately $2000\times$ faster.

Pretrained vs. single-image gap. The pretrained Neighbor2Neighbor achieves an average PSNR of 30.56 dB, a gain of 2.12 dB over the best single-image method (DIP). This gap represents the value of having learned a generalised natural image prior from a large training corpus. The SSIM gap is similarly pronounced (0.883 vs. 0.821), and edge preservation improves from 0.910 to 0.943.

Per-image variability. Image 0000 is a notable outlier: Self2Self achieves 36.48 dB on this image, its highest result by a wide margin, and the pretrained model achieves 38.41 dB. This suggests image 0000 has strong structural regularity particularly well-suited to both methods. Conversely, images 0007 and 0008 are consistently challenging for all methods, yielding PSNR values below 28 dB regardless of approach. Figure 1 shows the two images to illustrate the contrast in content complexity.

5 Related Works

Noise2Noise. Lehtinen et al. [2018] demonstrated that a neural network can be trained for denoising using pairs of independently corrupted observations of the same scene, without ever seeing a clean image. This foundational result motivated subsequent single-image approaches by showing that clean targets are not strictly necessary.

Blind-spot networks. Noise2Void [Krull et al., 2019] and Noise2Self [Batson and Royer, 2019] introduced the blind-spot paradigm, where the prediction for each pixel is structurally excluded from its own receptive field. This ensures the network cannot trivially learn the identity function on noisy inputs, providing a theoretically principled self-supervised loss without requiring paired data.

Recorrupted-to-Recorrupted (R2R). Pang et al. [2021] showed that adding fresh noise to an already noisy image and training on these re-corrupted pairs yields an unbiased estimator of the supervised loss. This approach extends single-image self-supervised denoising to settings where the noise distribution can be approximately characterised.

Iterative Data Refinement (IDR). Zhang et al. [2022] proposed iteratively refining the self-supervised training target by using the denoised output of one training round as the basis for the next. This progressive approach improves the quality of the implicit supervision signal over successive rounds.

Supervised methods. DnCNN [Zhang et al., 2017] and FFDNet [Zhang et al., 2018] established strong supervised baselines with residual learning on large datasets. While out of scope for this study (which focuses on self-supervised methods), they provide a ceiling reference: DnCNN achieves approximately 31.7 dB on CBSD68 at $\sigma = 25$, suggesting that the pretrained Neighbor2Neighbor (30.56 dB on our subset) approaches but does not yet match the best supervised methods.



(a) Image 0000 - the best-performing image across all methods (Self2Self: 36.48 dB, N2N pretrained: 38.41 dB). The image contains large smooth regions and repeating structures, providing strong spatial redundancy that self-supervised methods can exploit effectively.



(b) Image 0008 - the worst-performing images across all methods (best: DIP at 25.85 dB, N2N pretrained: 27.76 dB). The image contains complex, high-frequency detail with little spatial redundancy, making it difficult for patch-based or structure-exploiting priors to recover fine texture faithfully.

Figure 1: Qualitative comparison of the best-performing (left) and worst-performing (right) images in the evaluation set. Content characteristics — spatial regularity, texture complexity, and structural repetition — directly influence denoising difficulty across all methods.

6 Discussion

The results reveal several important observations about the current landscape of self-supervised single-image denoising.

The cost of operating without external data. The 2.12 dB PSNR gap between the pretrained Neighbor2Neighbor and the best single-image method (DIP) directly quantifies the value of a learned natural image prior. This gap is not attributable to model capacity — Self2Self and DIP use architectures of comparable complexity — but to the fundamental difference between inductive biases derived from the architecture alone versus those absorbed from millions of natural images. Closing this gap without external data remains an open problem.

PSNR vs. SSIM trade-off. DIP leads in PSNR while Self2Self leads in SSIM among single-image methods. This divergence suggests the two methods optimise for subtly different aspects of image quality. DIP’s pixel-level fidelity benefits from long per-image optimisation, while Self2Self’s dropout averaging may produce outputs with better-preserved

structural coherence even at slightly lower pixel accuracy.

Edge preservation as a complementary metric. Neighbor2Neighbor single achieves the highest edge preservation (0.930) despite ranking last in PSNR and SSIM. This suggests that sub-image pair training imposes a particularly effective constraint on structural boundaries, even when it does not globally minimise pixel error. Edge preservation captures a dimension of denoising quality not fully reflected by PSNR or SSIM.

Computational feasibility. All single-image methods are impractical for real-time or high-throughput applications. Self2Self’s relative efficiency (approximately half the time of DIP and Neighbor2Neighbor single) is an advantage, but the absolute cost remains high. Accelerating per-image optimisation — through better initialisation, adaptive stopping criteria, or architectural improvements — is an important practical direction.

References

- Joshua Batson and Loic Royer. Noise2Self: Blind denoising by self-supervision. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 524–533, 2019.
- Antoni Buades, Bartomeu Coll, and Jean-Michel Morel. A non-local algorithm for image denoising. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, volume 2, pages 60–65, 2005.
- Kostadin Dabov, Alessandro Foi, Vladimir Katkovnik, and Karen Egiazarian. Image denoising by sparse 3-D transform-domain collaborative filtering. *IEEE Transactions on Image Processing*, 16(8):2080–2095, 2007.
- Tao Huang, Songjiang Li, Xu Jia, Huchuan Lu, and Jianzhuang Liu. Neighbor2Neighbor: Self-supervised denoising from single noisy images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14781–14790, 2021.
- Alexander Krull, Tim-Oliver Buchholz, and Florian Jug. Noise2Void – learning denoising from single noisy images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2129–2137, 2019.
- Jaakko Lehtinen, Jacob Munkberg, Jon Hasselgren, Samuli Laine, Tero Karras, Timo Aila, and Timo Aila. Noise2Noise: Learning image restoration without clean data. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 2965–2974, 2018.
- David Martin, Charles Fowlkes, Doron Tal, and Jitendra Malik. The Berkeley segmentation dataset and benchmark. <https://www2.eecs.berkeley.edu/Research/Projects/CS/vision/bsds/>, 2001.
- Tongyao Pang, Huan Zheng, Yuhui Quan, and Hui Ji. Recorruped-to-recorruped: Unsupervised deep learning for image denoising. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2043–2052, 2021.
- Yuhui Quan, Mingqin Chen, Tongyao Pang, and Hui Ji. Self2Self with dropout: Learning

- self-supervised denoising from single image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1890–1898, 2020.
- Dmitry Ulyanov, Andrea Vedaldi, and Victor Lempitsky. Deep image prior. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9446–9454, 2018.
- Kai Zhang, Wangmeng Zuo, Yunjin Chen, Deyu Meng, and Lei Zhang. Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising. *IEEE Transactions on Image Processing*, 26(7):3142–3155, 2017.
- Kai Zhang, Wangmeng Zuo, and Lei Zhang. FFDNet: Toward a fast and flexible solution for CNN-based image denoising. *IEEE Transactions on Image Processing*, 27(9):4608–4622, 2018.
- Yi Zhang, Dasong Hua, Liwei Wang, Jianzhuang Liu, Kai Zhang, Zhida Li, and Bing Su. IDR: Self-supervised image denoising via iterative data refinement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17683–17692, 2022.